6	(a)	Magnetic flux is defined as the product of magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density. (from LO) Or Magnetic flux is the product of the flux density normal to the surface and the area of the surface. (from notes). [1]	
	(b)	(i)	90° or 270° [1]
		(ii)	Max Φ = NBA $= 300 \times 5.0 \times 10^{-2} \times \pi(4.0 \times 10^{-2})^2$ [C1] = 0.0754 Wb [A1]
		(iii)	Average emf $= \dfrac{\Delta\Phi}{\Delta t}$ $= \dfrac{(0.0754-0)}{0.50\times10^{-3}}$ [C1] $= 151\,\text{V}$ [A1]
		(iv)	The minimum e.m.f. can be determined from the time at which the gradient of the graph is zero. [1] The first instant is at t = 0.50 ms. (no marks deducted for sf) [1]
		(v)	As the coil rotates, the magnetic flux linkage of the coil varies sinusoidally. By Faraday's Law, the induced e.m.f is directly proportional to the rate of change of magnetic flux linkage. [M1] Hence the e.m.f will also vary sinusoidally. [A1]